Exploring the Solar System in 3D

Solar System Scope · Solar System Scope

Years 7, 9-10	~30 minutes	Science · Earth & Space Science

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This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://www.solarsystemscope.com/
Curriculum links: AC9S7U03 · Earth geometry · Earth systems

Learning intentions

- I can describe how the Earth, sun and moon move relative to each other and how this creates day/night, seasons and eclipses.
- I can compare the relative size, distance and orbital speed of planets in the solar system.
- I can use a 3D model to explain patterns in planetary motion.

Getting started

Open Solar System Scope and use the mouse or touch controls to fly around the solar system, zooming in on Earth first. Use the time controls to speed up time and watch the Earth, moon and sun move relative to each other, then fly out to compare Earth with the other planets.

Tasks

1. Zoom in on the Earth-moon system and speed up time. Describe what you observe about how the moon moves around the Earth while the Earth spins.

2. Find a position where the sun, Earth and moon roughly line up. What natural event could occur when they line up this way, and why?

3. Speed up time over about a year (in simulation time) while watching Earth's orbit and its tilt. Explain, based on what you observe, why Australia and countries in the Northern Hemisphere have opposite seasons at the same time of year.

4. Fly out to view the whole solar system. Choose four planets and record their approximate distance from the Sun and how long each takes to complete one orbit, using the simulation's information panels.

Planet	Approx. distance from Sun	Approx. time to orbit once

5. Based on your table, describe the relationship between a planet's distance from the Sun and how long it takes to orbit.

6. Compare the size of Earth with the size of Jupiter or Saturn by flying between them. Estimate roughly how many Earths could fit across the diameter of the larger planet, and explain how you made your estimate.

7. Switch to night-sky mode and set the date to today. Identify one constellation or bright object visible from your location tonight.

8. Change the date in night-sky mode to six months from today. Explain why the visible constellations change over the course of a year, connecting your answer to Earth's motion.

Extension (optional)

Use the simulation to compare the orbital speed of a planet close to the Sun (like Mercury) with one far away (like Neptune). Propose an explanation for why planets closer to the Sun orbit faster, and describe what evidence from the simulation supports your idea.